

PRAKTIKUM SISTEM OPERASI

MODUL 10

Shell



Oleh:

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2311104072

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

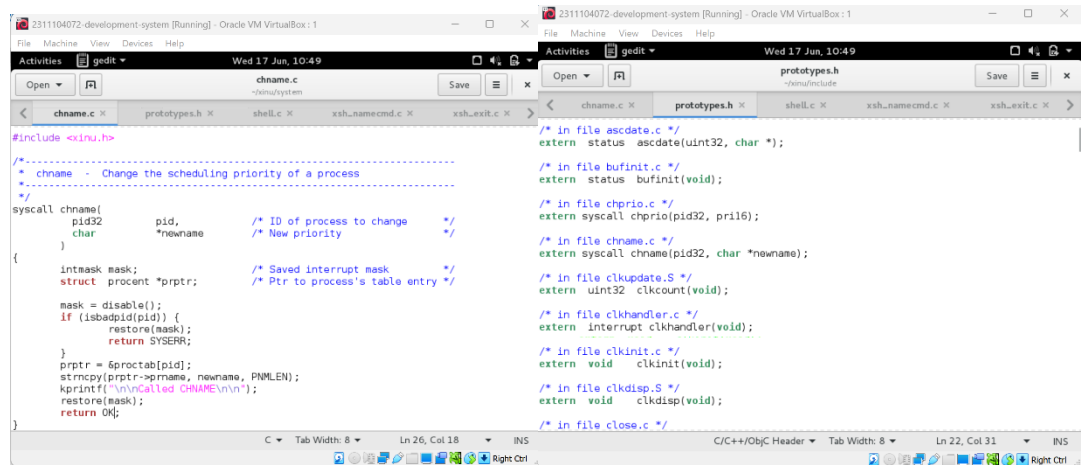
DIREKTORAT KAMPUS PURWOKERTO

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JURNAL

1. Soal 1



The image shows two side-by-side Oracle VM VirtualBox windows, each running a development system. The left window displays the `chname.c` source file, which implements a `syscall` named `chname` to change the scheduling priority of a process. It includes `<linux.h>` and uses `prctl` to set the priority. The right window displays the `prototypes.h` header file, which defines the `chname` function and lists various other system call prototypes like `ascdade`, `bufinit`, `chprio`, `clkupdate`, `clkhandler`, `clkinit`, `clkdisp`, and `close`.

```
#include <linux.h>

/*-----
 * chname - Change the scheduling priority of a process
 *-----
 */

syscall chname(
    pid32      pid,      /* ID of process to change */
    char       *newname  /* New priority */
)
{
    intmask mask;        /* Saved interrupt mask */
    struct procent *prptr; /* Ptr to process's table entry */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    strcpy(prptr->prname, newname, PNMLEN);
    kprintf("\nvcalled CHNAME\n");
    restore(mask);
    return OK;
}
```

```
/* in file ascdade.c */
extern status ascdade(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern syscall chname(pid32, char *newname);

/* in file clkupdate.S */
extern uint32 clkcount(void);

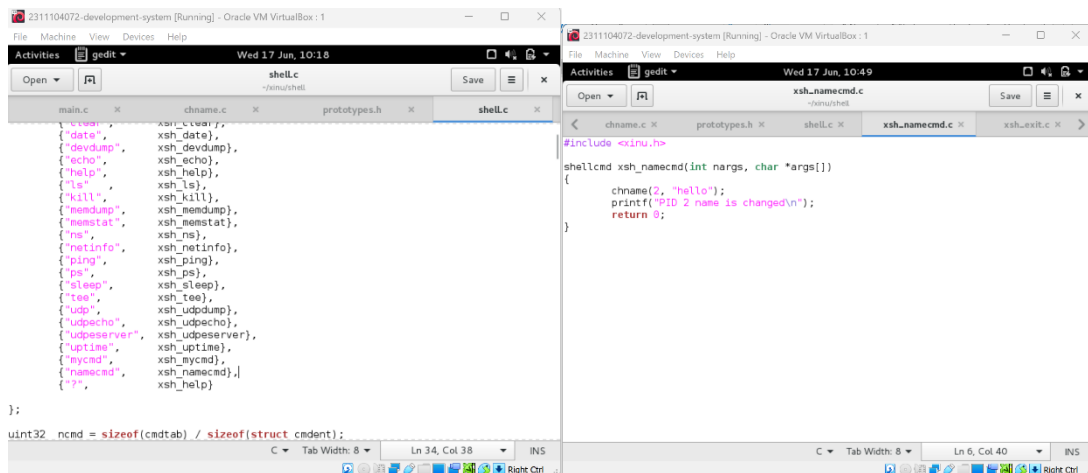
/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */
```

2. Soal 2



The image shows two side-by-side Oracle VM VirtualBox windows. The left window displays the `main.c` file, which defines a table of shell commands and their corresponding functions. The right window displays the `xsh_namecmd.c` file, which implements the `shellcmd` function for the `xsh_namecmd` command.

```
main.c
{
    {"date", xsh_date},
    {"devdump", xsh_devdump},
    {"echo", xsh_echo},
    {"help", xsh_help},
    {"ls", xsh_ls},
    {"kill", xsh_kill},
    {"mendump", xsh_mendump},
    {"menstat", xsh_menstat},
    {"ns", xsh_ns},
    {"netinfo", xsh_netinfo},
    {"ping", xsh_ping},
    {"ps", xsh_ps},
    {"sleep", xsh_sleep},
    {"tee", xsh_tee},
    {"udp", xsh_udpdump},
    {"udpecho", xsh_udpecho},
    {"udpservice", xsh_udpservice},
    {"uptime", xsh_uptime},
    {"mycmd", xsh_mycmd},
    {"namecmd", xsh_namecmd},
    {"?", xsh_help}
};

uint32 ncmd = sizeof(cmdtab) / sizeof(struct cmdent);
```

```
xsh_namecmd.c
#include <linux.h>

shellcmd xsh_namecmd(int nargs, char *args[])
{
    chname(2, "hello");
    printf("ID 2 name is changed\n");
    return 0;
}
```

3. Soal 3

```
2311104072-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Wed 17 Jun, 10:51
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/include
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005F7FFC 0x005F794C 8192 0 0
6 ps         curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0
ksh $ namecmd

Called CHNAME
PID 2 name is changed
ksh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 hello       wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005F7FFC 0x005F794C 8192 0 0
6 ps         curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0
ksh $
Ctrl-A Z for help | 115200 8NI | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```